Chapter 6

Conclusion

6.1. Summary of the Study

This study set out to address a clearly identified and practically important gap in the research landscape of Bangladesh: the absence of any large-scale, algorithmically rigorous analysis of suicide patterns from the country's most comprehensive publicly available data source — newspaper-reported cases. The core challenge was not merely analytical but structural: newspaper-based datasets from Bangladesh are characterised by extreme and non-random missingness, often exceeding 50–90% for several important features, making conventional machine learning pipelines unreliable without careful pre-treatment of the data.

To address this, the study built and executed a comprehensive end-to-end research framework spanning dataset construction, preprocessing, imputation, clustering, and evaluation. The original Kaggle dataset of 760 records was substantially expanded through manual collection of 857 additional cases from verified Bangladeshi news sources, yielding a final analytical dataset of 1,617 records and 34 columns covering the period 2017–2025. Nine new socioeconomic and clinical attributes were added, and missingness in spatial features was dramatically reduced through the application of geocoding (Geopy) and meteorological (Meteostat) APIs.

Three state-of-the-art imputation methods — MICE, KNN, and MissForest — were each applied under two predictor-selection strategies (context-aware and no-context), generating six fully imputed datasets alongside two non-imputed baselines. Four established clustering algorithms — K-Means, Agglomerative Clustering, DBSCAN, and Spectral Clustering — were then applied across all eight datasets, yielding 32 complete imputation–clustering pipeline combinations. All 32 pipelines were evaluated using five standard internal clustering validity indices supplemented by a newly proposed balance metric that penalises cluster size imbalance and excessive noise assignment — a methodological contribution designed specifically for the natural imbalances encountered in social datasets of this kind.

The best-performing pipeline — Agglomerative Clustering with MissForest no-context imputation — achieved a composite score of 27.30 out of 32. Its two-cluster solution was among the most practically meaningful results of the entire study, and is discussed in detail in the following section.

6.2. Key Findings

The study produced findings across four distinct dimensions: methodological, empirical, evaluative, and practical.

Finding 1 — MissForest consistently outperforms MICE and KNN on this data type: Across both predictor strategies and across all four clustering algorithms, MissForest-imputed datasets produced more consistent, better-ranked clustering solutions than either MICE or KNN. This is consistent with the international literature on imputation benchmarks [11, 23, 24] and confirms that MissForest's ensemble-based, non-parametric approach is particularly well-suited to the mixed-type, high-missingness characteristics of newspaper-derived social data.

Finding 2 — Imputed datasets vastly outperform non-imputed baselines: Both non-imputed baseline datasets (raw and minimal-fill) ranked at or near the bottom of the composite scoring table. This confirms that attempting to cluster severely incomplete data without principled imputation produces structurally weak and practically uninterpretable results — a finding with direct implications for future researchers working with similarly incomplete datasets in LMICs.

Finding 3 — The balance metric was essential for fair evaluation: Several DBSCAN configurations produced deceptively high Silhouette scores and Calinski–Harabasz values while simultaneously assigning over 99% of data points as noise, leaving fewer than 20 observations in meaningful clusters. Without the balance metric — weighted at 35% of the composite score — these configurations would have ranked among the top solutions. This finding highlights a critical and underappreciated limitation of standard clustering validity indices when applied to naturally imbalanced real-world datasets [29, 30].

Finding 4 — The best pipeline produced a two-cluster solution with strong structural and temporal interpretability: The most significant and arguably most insightful finding of the study is the nature of the two-cluster solution produced by the best-performing pipeline (agg_rf_no_context). Cluster analysis revealed a clear, coherent separation in the data that aligned directly with the dataset's construction history: cases from the original 2020-focused Kaggle dataset formed one cluster, while the newly collected cases covering 2017–2019 and 2021–2025 (i.e., the years before and after 2020) formed the other.

This separation was not imposed by design — the clustering algorithm had no access to collection year, data source, or any direct temporal identifier during the clustering process. The fact that it emerged organically from the feature-level patterns in the data is a powerful validation of the pipeline's quality. It demonstrates that the newly collected cases, which were gathered independently and contain systematically higher missingness rates across several columns, carry a distinct and detectable multivariate signature that the MissForest imputation preserved faithfully rather than erased through over-smoothing.

This finding has two important implications. First, it confirms that the imputation strategy successfully maintained the structural differences between the two data collection cohorts rather than artificially homogenising them — which is precisely what a good imputation method should do. Second, it suggests that missingness itself carries meaningful information about the data-generating process: cases with higher missingness are not simply records with absent values, but reflect a distinct reporting environment characterised by less consistent newspaper coverage, different geographic distribution, and different temporal patterns compared to the more densely covered 2020 cases.

Finding 5 — Spectral Clustering showed the strongest average algorithmic performance: Across all datasets and composite metric components, Spectral Clustering achieved the highest average algorithm-level performance, though its computational cost and sensitivity to parameter choices made it less practically reliable than Agglomerative Clustering for this specific dataset. Agglomerative Clustering combined the second-best algorithmic performance with greater stability across imputation variants, making it the most robust choice for this type of social dataset.

6.3. Achievement of Research Objectives

All six research objectives defined in Chapter 1 were fully achieved within the scope of this study. The following table summarises each objective against the corresponding achievement:

Objective	Achievement	Status
Expand and enrich dataset from 760 to 1,617 records with 9 new attributes covering 2017–2025	Dataset expanded to 1,617 records and 34 analytical columns; 9 new socioeconomic and clinical attributes added	Fully Achieved
Reduce missingness in spatial and weather features using Geopy and Meteostat APIs	Latitude/longitude missingness reduced from ~53% to 6.49%; partial reduction in weather variables achieved	Fully Achieved
Implement and compare MICE, KNN, and MissForest under context-aware and no-context predictor strategies	Six imputed datasets produced across three methods and two predictor strategies; all post-imputation validation checks passed	Fully Achieved
Apply four clustering algorithms across all 8 datasets generating 32 pipeline combinations	All 32 pipelines executed and evaluated; results stored and ranked	Fully Achieved
Evaluate all 32 pipelines using five standard validity indices supplemented by a novel balance metric	Composite scoring framework implemented; balance metric introduced and weighted at 35% of total composite score	Fully Achieved
Identify the best pipeline and provide detailed interpretable cluster profiles	Best pipeline identified (agg_rf_no_context, composite score 27.30/32); two-cluster solution interpreted across demographic, geographic, and temporal dimensions	Fully Achieved

Table 6.1: Research objectives and their achievement status

It is worth noting that the achievement of Objective 6 — the cluster profiling — was more nuanced than a simple binary outcome. The two-cluster solution produced by the best pipeline was fully interpretable and yielded meaningful insights, but the nature of the separation (data-cohort-driven rather than purely sociological) means that richer demographic profiling of within-cohort subgroups remains a natural direction for future work, as discussed in Section 6.5.

6.4. Limitations of the Study

Despite the methodological rigour of the framework, the study has several important limitations that must be acknowledged:

Limitation 1 — Data source bias: The dataset is drawn exclusively from newspaper-reported completed suicides. Newspaper coverage is inherently biased towards cases that are deemed newsworthy — typically those involving unusual methods, public locations, or prominent individuals — and systematically under-represents remote rural suicides and cases involving stigmatised populations. The findings therefore reflect patterns within reported cases rather than the true epidemiological distribution of suicide in Bangladesh.

Limitation 2 — Limited dataset size: Although 1,617 records represents the largest manually curated newspaper-based Bangladeshi suicide dataset to date, it remains relatively small by machine learning standards. High missingness in many features further reduces the effective information content of the dataset, limiting the resolution of cluster boundaries and the statistical power of within-cluster analyses. A larger dataset would allow for the exploration of more granular clustering solutions with three or more clusters.

Limitation 3 — Pipeline search space: While 32 pipelines represent a comprehensive comparison by the standards of this literature, the study did not exhaustively explore all possible hyperparameter combinations within each algorithm. For example, only a subset of DBSCAN epsilon and minimum-samples combinations were tested; and although K-Means and Agglomerative Clustering were searched over k=2–20 and k=2–10 respectively, k=2 consistently emerged as the optimal cluster count for the best-performing imputed pipelines, leaving higher-k structures comparatively under-explored at the interpretation stage. A more exhaustive grid search across a wider hyperparameter space might identify solutions with different cluster structures that offer additional interpretive value.

Limitation 4 — No external validation: Because the study is entirely unsupervised, there is no ground-truth label set against which to validate the cluster assignments. While the temporal coherence of the best-pipeline clusters provides strong indirect validation, external validation using independently collected data — such as clinical records or police reports — would provide a more rigorous test of whether the identified clusters correspond to genuinely distinct epidemiological subgroups.

Limitation 5 — Exclusion of Bangla free-text columns: Several potentially highly informative columns — including profession_description and reason_description — contained free-text entries in Bangla and could not be processed with standard NLP tools available within the project's resource constraints. These columns were excluded from the analysis. Their inclusion through Bangla-specific natural language processing could substantially enrich the feature space and enable more semantically nuanced cluster differentiation.

Limitation 6 — Incomplete weather data: Despite the use of the Meteostat API, weather-related variables remained substantially incomplete for many rural locations due to sparse meteorological station coverage in those areas. The contribution of weather and environmental factors to the cluster structure is therefore likely underestimated in the current analysis.

6.5. Directions for Future Work

The findings and limitations of this study point towards several promising and important directions for future research:

1. Understanding what drives the cluster separation — and using it for richer profiling: The most immediate and impactful next step is to investigate more deeply what specific combinations of features — beyond data-collection cohort membership — drive the two-cluster separation identified by the best pipeline. By systematically analysing which features contribute most to the inter-cluster distance (through techniques such as feature importance from random forests trained on cluster labels, or SHAP-based explainability), future work can determine whether the separation reflects genuine sociological differences in suicide patterns across different time periods, or whether it is primarily an artefact of differential reporting practices. If genuine temporal and demographic shifts can be isolated from reporting artefacts, the findings could directly inform how suicide prevention strategies should be adapted as Bangladesh's social and economic landscape continues to evolve.

2. Collecting more data and expanding temporal coverage: The current dataset covers 2017–2025 but with substantially uneven density across years. A sustained data collection effort that adds cases from vernacular Bangla newspapers — currently under-represented — and that populates missing attributes more consistently would enable exploration of three or more meaningful cluster solutions, potentially revealing distinct risk profiles for student suicides, rural male suicides, and urban domestic-conflict-related suicides as separate, interpretable groups rather than as within-cluster variations.

3. Incorporating Bangla NLP for free-text feature extraction: The exclusion of Bangla free-text columns represents a significant loss of information. Future work should apply Bangla-specific natural language processing — for example using BanglaBERT [46] or similar transformer-based models fine-tuned on Bangla social text — to extract structured features from reason_description and profession_description fields. These features could substantially improve the semantic richness of the feature space and enable the discovery of more nuanced patterns related to motivation, occupation, and social context.

4. Moving from pattern discovery to causal understanding: The present study identifies that distinct groups exist within the data — but it does not explain why those groups are structured as they are, or which factors are causally linked to elevated suicide

risk within each group. Future work could apply causal inference methods [47] — such as propensity score matching or structural equation modelling — to the cluster-labelled dataset to test hypotheses about the causal pathways from socioeconomic stressors, seasonal factors, and demographic characteristics to suicide risk. This transition from descriptive clustering to causal modelling would represent a major step towards generating actionable, evidence-based prevention recommendations.

5. Developing a targeted early-warning framework: In the longer term, the cluster profiles identified in this study could serve as the foundation for a targeted early-warning or risk-stratification framework for suicide prevention in Bangladesh. If the distinguishing features of high-risk clusters — particularly those related to geography, time of year, occupation, and family circumstances — can be monitored through routinely available administrative or news data, it may be possible to design a lightweight, near-real-time alerting system that flags geographic areas or time periods of elevated risk to mental health services and NGOs. This kind of applied extension would directly fulfil the beneficence objective of the research and align with the growing literature on AI-assisted suicide prevention [41, 42].

6. Extending the framework to other LMICs: The methodological pipeline developed in this study — combining API-based conditional filling, multi-method imputation comparison, balance-aware composite evaluation, and interpretable cluster profiling — is not specific to Bangladesh or to suicide data. It is directly applicable to any severely incomplete, newspaper-derived social dataset in a low- or middle-income country context. Future work could validate and extend the framework to comparable datasets from South Asian or Sub-Saharan African countries where media-based data collection faces similar structural challenges [22], contributing to the development of a generalisable toolkit for pattern discovery from incomplete social data in resource-constrained settings.

6.6. Concluding Remarks

This study has demonstrated that meaningful, structurally coherent, and practically interpretable patterns can be recovered from severely incomplete, newspaper-derived social data — even when missingness exceeds 90% for several key features — provided that the right combination of imputation strategy, clustering algorithm, and evaluation framework is applied. The systematic comparison of 32 imputation–clustering pipelines, evaluated through a balance-aware composite scoring framework, provides the most rigorous methodological treatment of this type of data yet attempted in the Bangladeshi suicide research literature.

The two-cluster solution produced by the best pipeline — Agglomerative Clustering with MissForest no-context imputation — is particularly noteworthy because the cluster separation emerged organically without any temporal or source-based supervision. This is not merely a technical result: it is a demonstration that carefully designed imputation can preserve genuine structural differences in data rather than artificially homogenising them, and that clustering can detect those differences even when they are subtle and embedded within a high-missingness feature space. The finding that the cluster boundary aligns with the data collection cohort boundary — and therefore reflects real differences in reporting patterns, geographic coverage, and temporal context — is a valuable insight into the data-generating process itself.

Ultimately, this research is motivated not by methodological novelty alone, but by the hope that better analytical tools applied to existing — however imperfect — data sources can contribute, however modestly, to a more evidence-based approach to suicide prevention in Bangladesh. Every cluster profile, every geographic risk pattern, and every temporal trend identified in this study represents a potential target for intervention. The 10,000 or more lives lost to suicide in Bangladesh every year [1, 2] represent not just a public health statistic, but a call for precisely the kind of rigorous, data-driven, and ethically grounded research that this study has attempted to provide.